SECTION A

(25 *marks*)

1. Solve the following :

(a) $\lim_{t\to 1}\dfrac{t^3+t^2-5t+3}{t^3-3t+2}$ **(4 marks)**

(b) $\lim_{x\to 0}\dfrac{x}{\sqrt{1+x+x^2}-1}$ **(5 marks)**

2. **(a)** By using the definition of $f(x)'=\lim_{x\to a}\dfrac{f(x)-f(a)}{x-a}$, find the values of a and b such that $f\,'(1)$ exist if

$$f(x)=\begin{cases}|x-1| \ , & x<1\\ ax+b \ , & x\geq 1\end{cases}$$ **(6 marks)**

(b) Find the derivative of the following functions

$$y=\frac{\sin x}{1-\cos x}$$ **(4 marks)**

3. A closed right circular cylindrical container of radius r and height h is to be constructed with volume 4,000cm^3. The cost for the construction is RM1.00 per cm^2 for the curved surface while RM2.00 per cm^2 for the top and bottom surfaces.

State h in terms of r and hence, find the radius of the cylinder so that the cost of the construction is minimum. **(6 marks)**

SECTION B

(75 *marks*)

1. Given the complex number $z=\dfrac{4}{i-\sqrt{3}}$. Find the modulus and argument of z. Express z in the form $r\left(\cos\theta+i\sin\theta\right)$. **(5 marks)**

2. **(a)** Solve $\log_p\left(5+x^2\right)=\log_{\sqrt{p}}\left(3-x\right)$. **(6 marks)**

(b) Find the solution set of the inequality $\dfrac{x+8}{3-x}\leq 4$. **(6 marks)**

3. Given $f(x)=\sin(2x-\pi)$ for $0\leq x\leq 2\pi$.

(a) Sketch the graph of $f(x)$. **(3 marks)**

(b) Find the domain and range of $f(x)$. (**2 marks**)

4. Function f and g are defined as follow:

$$f(x)=x^2-8x+15, \qquad x\leq 4$$

$$g(x)=4+\ln x, \qquad x\geq 0$$

(a) Find $f^{-1}(x)$, $g^{-1}(x)$ and their domain. **(7 marks)**

(b) Hence, find $\left(f\circ g\right)^{-1}\left(x\right)$. **(3 marks)**

(c) Find $h(x)$ where $\left(h\circ g\right)\left(x\right)=\ln x^2+1$. **(4 marks)**

5. Solve the following:

(a) Determine the continuity of $f(x)$ at $x=1$.

$$f(x)=\begin{cases} \dfrac{x^2+x-2}{x-1} & , \quad x>1 \\ 1 & , \quad x=1 \\ x^3+2 & , \quad x<1 \end{cases}$$

(5 marks)

(b) Find the vertical and horizontal asymptotes for graph of $f(x)=\dfrac{x^2+1}{x-1}$. Hence sketch the graph of $f(x)$. **(9 marks)**

6. The curve with equation $y=\dfrac{a+4x-x^2}{2-ax-x^2}$, where a is constant, passes through the point (1,1).

(a) Calculate the value of a. **(2 marks)**

(b) Hence, find all the asymptotes of the curve. **(4 marks)**

(c) Express y in partial fractions. Hence, find the values of $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$ when $x = 0$.

(8 marks)

7. **(a)** Find the local minimum and the local maximum of the curve $f(x)=x+\dfrac{9}{x}$.

(5 marks)

(b) Water is poured into vase at a constant rate of $80\pi cm^3 s^{-1}$. When the depth of water is h cm, the volume of water, V cm^3 is given by $V=4\pi h(h+4)$. Find the rate of change of the depth of water in the vase when $h=6$cm. **(5 marks)**

END OF QUESTIONS PAPER

ANSWER

SECTION A

1. (a) $\frac{4}{3}$ (b) 2

2. (a) a = -1, b =1 (b) $\frac{-1}{1-\cos x}$

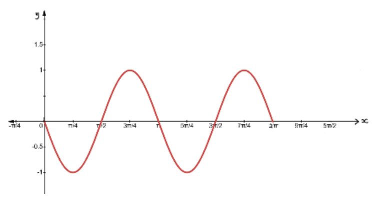
3. $h=\frac{4000}{\pi r^2}$, r = 6.83

SECTION B

1. $|z|=2,\ Arg(z)=-2.618$ rad or $-\frac{5}{6}\pi$ rad $z=2\left[\cos(-2.618)+i\sin(-2.618)\right]$

2. (a) $x=\frac{2}{3}$ (b) $\left\{x : x\le\frac{4}{5} \text{ or } x>3\right\}$

3. (a)



(b) $D_f=[0,\ 2\pi],\ \ R_f=[-1,\ 1]$

4. (a) $f^{-1}(x)=4-\sqrt{x+1}$, $D_{f^{-1}}=[-1,\infty]$

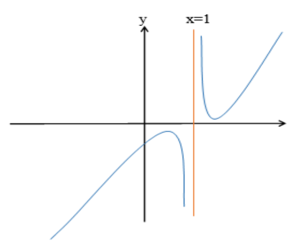
$g^{-1}(x)=e^{x-4},\quad D_{g^{-1}}=\mathbb{R}$

(b) $e^{-\sqrt{x+1}}$ (c) $h(x)=2x-7$

5. (a) $f(x)$ discontinuous at $x=1$

(b) The vertical asymptote is $x=1$

No horizontal asymptote for $f(x)$



6. (a) a = -1 (b) V.A= 2, -1, H.A y=1

(c) $\frac{-1+4x-x^2}{2+x-x^2}=1-\frac{2}{1+x}+\frac{1}{2-x}$. $\frac{dy}{dx}=\frac{9}{4}$. $\frac{d^2y}{dx^2}=\frac{-15}{4}$

7. (a) $(3,6)$ minimum, $(-3,-6)$ maximum (b) $\frac{5}{4}\ cms^{-1}$

SECTION A

(25 *marks*)

NO	ANSWER SCHEMES	REMARKS
1(a)	$\lim_{t\to 1}\frac{t^3+t^2-5t+3}{t^3-3t+2}$	**K1** **(Both Factorize)**
	$\lim_{t\to 1}\frac{(t-1)^2(t+3)}{(t-1)^2(t+2)}$	**K1** **(Simplify)**
	$\lim_{t\to 1}\frac{(t+3)}{(t+2)}$	**K1J1**
	$\frac{1+3}{1+2}=\frac{4}{3}$	
(b)	$\lim_{x\to 0}\frac{x}{\sqrt{1+x+x^2}-1}$	
	$\lim_{x\to 0}\frac{x}{\sqrt{1+x+x^2}-1}\times\frac{\sqrt{1+x+x^2}+1}{\sqrt{1+x+x^2}+1}$	**K1** **(Multiply conjugate)**
	$\lim_{x\to 0}\frac{x\left(\sqrt{1+x+x^2}+1\right)}{\left(1+x+x^2\right)-1}$	**K1** **(Simplify)**
	$\lim_{x\to 0}\frac{x\left(\sqrt{1+x+x^2}+1\right)}{x(1+x)}$	**K1**
	$\lim_{x\to 0}\frac{\left(\sqrt{1+x+x^2}+1\right)}{(1+x)}$	**K1**
	$\frac{\left(\sqrt{1+0+0^2}+1\right)}{(1+0)}=2$	**J1**
	TOTAL	**9 MARKS**

NO	ANSWER SCHEMES	REMARKS
2(a)	$f'(1)=\lim_{x\to1^-}\frac{1-x-(0)}{x-1}=\lim_{x\to1^+}\frac{ax+b-(a+b)}{x-1}$ $\lim_{x\to1^-}\frac{1-x}{x-1}=\lim_{x\to1^+}\frac{ax-a}{x-1}$ $\lim_{x\to1^-}-1=\lim_{x\to1^+}a$ $a=-1$ f is differentiable at x = 1. Hence, a = -1 f is continuous at x = 1 if $\lim_{x\to1^-}f(x)=\lim_{x\to1^+}f(x)$ $\lim_{x\to1^-}(1-x)=\lim_{x\to1^+}(ax+b)$ $0=\lim_{x\to1^+}(-x+b)$ $-1+b=0$ $b=1$	**K1K1** **J1** **K1K1** **J1**
2(b)	$y=\frac{\sin x}{1-\cos x}$ $u=\sin x \quad v=1-\cos x$ $u'=\cos x \quad v'=\sin x$ $\frac{dy}{dx}=\frac{(1-\cos x)(\cos x)-\sin x(\sin x)}{(1-\cos x)^2}$ $=\frac{\cos x-\cos^2 x-\sin^2 x}{(1-\cos x)^2}$ $=\frac{\cos x-(\cos^2 x+\sin^2 x)}{(1-\cos x)^2}$ $=\frac{\cos x-1}{(1-\cos x)^2}$ $=\frac{-1}{1-\cos x}$	**K1** (correct differentiate of u & v) **K1** (use formula of quotient correctly) **K1** (use identity trigonometry) **J1**
	TOTAL	**10 MARKS**

NO	ANSWER SCHEMES	REMARKS
3	$V = 4000cm^3$	
	$V = \pi r^2 h = 4000cm^3, h = \dfrac{4000}{\pi r^2}$................(1)	B1
	Area of cylinder, $A =$(top and bottom) + curve surface $A = 2\pi r^2 + 2\pi rh$	K1
	Cost of the cylinder, $C =$RM2$(2\pi \text{r}^2)$ + RM1$\left(2\pi \text{r}h\right)$ $4\pi r^2 + 2\pi \text{rh}$.................(2)	K1
	Substitute (1) into (2): $C = 4\pi \text{r}^2 + 2\pi \text{r}\left(\dfrac{4000}{\pi r^2}\right)$ $= 4\pi \text{r}^2 + 8000r^{-1}$	K1
	$\dfrac{dc}{dr} = -8000r^{-2} + 8\pi r$ when $\dfrac{dc}{dr} = 0 \Rightarrow -8000r^{-2} + 8\pi r = 0$ $r = \sqrt[3]{\dfrac{1000}{\pi}} = 6.83$	
	$\dfrac{d^2c}{dr^2} = 16000r^{-3} + 8\pi$	K1
	when $r = 6.83$, $\dfrac{d^2c}{dr^2} = 16000(6.83)^{-3} + 8\pi = 75.35 > 0$ (**Minimum**) Cost is minimum when $\dfrac{d^2c}{dr^2} > 0$ $\therefore r = 6.83$	J1
	TOTAL	**6 MARKS**

SECTION B
(*75 marks*)

NO	ANSWER SCHEMES	REMARKS
1	$z=\dfrac{4}{-\sqrt{3}+i}\times\dfrac{-\sqrt{3}-i}{-\sqrt{3}-i}$	**K1** (multiply with conjugate)
	$=\dfrac{-4\sqrt{3}-4i}{3+1}$	
	$=-\sqrt{3}-i$	**J1**
	$\lvert z\rvert=\sqrt{\left(-\sqrt{3}\right)^2+\left(-1\right)^2}=2$	**K1** (correct formula modulus)
	$z=-\sqrt{3}-i$	
	$\theta=-\left(\pi-\tan^{-1}\left(\left\lvert\dfrac{-1}{-\sqrt{3}}\right\rvert\right)\right)$	
	$=-\left(\pi-0.5236\right)$	**K1** (correct formula argument)
	$=-2.618$ rad or $-\dfrac{5}{6}\pi$ rad	
	$z=2\left[\cos\left(-2.618rad\right)+i\sin\left(-2.618rad\right)\right]$	**J1**
	TOTAL	**5 MARKS**

NO	ANSWER SCHEMES	REMARKS
2(a)	$\log_p\left(5+x^2\right)=\log_{\sqrt{p}}\left(3-x\right)$	
	$\log_p\left(5+x^2\right)=\dfrac{\log_p\left(3-x\right)}{\log_p\sqrt{p}}$	**K1** (correct formula change base)
	$\log_p\left(5+x^2\right)=\dfrac{\log_p\left(3-x\right)}{\log_p p^{\frac{1}{2}}}$	**K1** (apply law of log)
	$\log_p\left(5+x^2\right)=2\log_p\left(3-x\right)$	**K1** (simplify)
	$\log_p\left(5+x^2\right)=\log_p\left(3-x\right)^2$	
	$\left(5+x^2\right)=\left(3-x\right)^2$	**K1** (compare both sides)
	$5+x^2=9-6x+x^2$	
	$x=\dfrac{2}{3}$	**K1** (expand and simplify)
		J1

NO	ANSWER SCHEMES	REMARKS
2(b)	$\frac{x+8}{3-x} \le 4$ $\frac{x+8}{3-x} - 4 \le 0$ $\frac{x+8-4(3-x)}{3-x} \le 0$ $\frac{5x-4}{3-x} \le 0$	**K1** (bring all to left side) **K1** (equate denominator **K1** (simplify)

	$\left(-\infty, \frac{4}{5}\right]$	$\left[\frac{4}{5}, 3\right)$	$(3, \infty)$
$5x-4$	$-$	$+$	$+$
$3-x$	$+$	$+$	$-$
$\frac{5x-4}{3-x}$	$-$	$+$	$-$

K1 (Using table of signs / positive number lines)

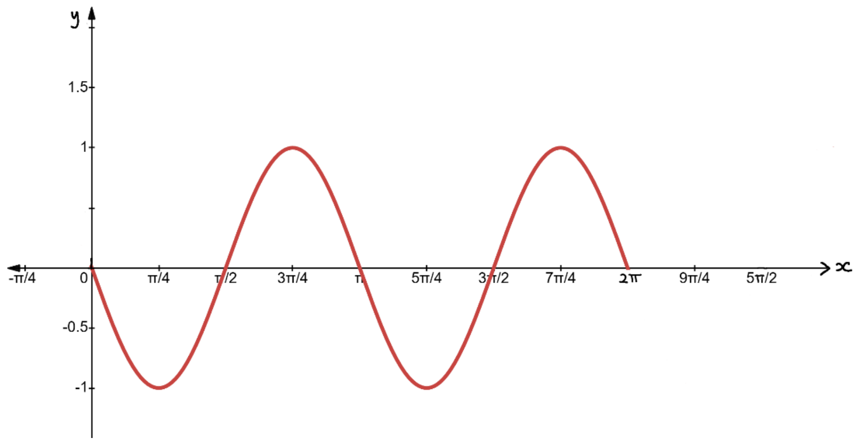
J1 (all correct)

Answer: $\left\{x : x \le \frac{4}{5} \text{ or } x > 3\right\}$ **J1**

	TOTAL	12 MARKS
NO	ANSWER SCHEMES	REMARKS
3 (a)	$f(x) = \sin(2x - \pi)$ $2x - \pi = 0 \Rightarrow x = \frac{\pi}{2} \quad \therefore \alpha = \frac{\pi}{2}$ $Start: \quad 0 + \frac{\pi}{2} = \frac{\pi}{2}$ $End: \quad 2\pi + \frac{\pi}{2} = \frac{5\pi}{2}$ $Step: \quad \frac{2\pi}{4(2)} = \frac{\pi}{4}$	

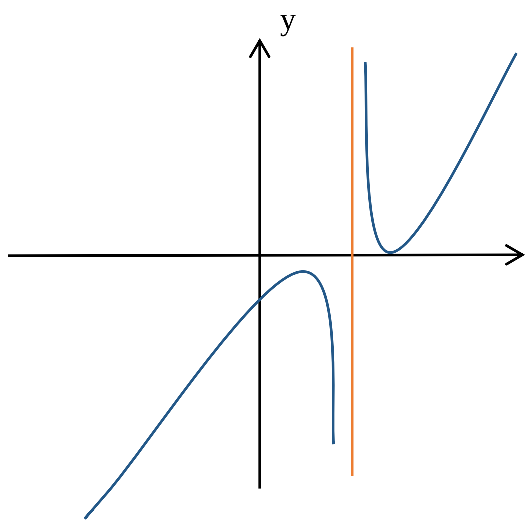
x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π	$\frac{9\pi}{4}$	$\frac{5\pi}{2}$
y	0	-1	0	1	0	-1	0	1	0	-1	0

K1

		**R1** **R1**
3(b)	$D_f=[0,\ 2\pi],\quad R_f=[-1,\ 1]$	**J1 J1**
	TOTAL	**5 MARKS**
NO	**ANSWER SCHEMES**	**REMARKS**
4 (a)	$f(x)=x^2-8x+15$ $\quad =(x-4)^2-1$ $f\left[f^{-1}(x)\right]=x$ $\left(f^{-1}(x)-4\right)^2-1=x$ $\left(f^{-1}(x)-4\right)^2=x+1$ $\left(f^{-1}(x)-4\right)=\pm\sqrt{x+1}$ Since $x\le 4$, $\quad f^{-1}(x)=4-\sqrt{x+1}$ $D_{f^{-1}}=[-1,\infty]$ $g\left[g^{-1}(x)\right]=x$ $4+\ln\left[g^{-1}(x)\right]=x$ $g^{-1}(x)=e^{x-4}$ $D_{g^{-1}}=\mathbb{R}$	**K1** (cts) **K1** (find inverse f) **J1** (ans f^{-1}) **J1** (domain f inverse) **K1** (find g^{-1}) **J1** (ans g^{-1}) **J1** (domain g inverse)
4(b)	Using properties $(f\circ g)^{-1}=g^{-1}\circ f^{-1}$. $(f\circ g)^{-1}=g^{-1}\circ f^{-1}=g^{-1}\left(4-\sqrt{x+1}\right)$ $\quad =e^{\left(4-\sqrt{x+1}\right)-4}$ $\quad =e^{-\sqrt{x+1}}$	**K1** (use property) **K1** **J1**

4(c)	$h \circ g = \ln x^2 + 1$ $h(4+\ln x) = \ln x^2 + 1$ $Let\, u = 4 + \ln x \quad \Rightarrow \quad \ln x = u - 4$ $h(4+\ln x) = 2\ln x + 1$ $h(u) = 2(u-4)+1$ $= 2u - 7$ $\therefore h(x) = 2x - 7$	**K1** **K1** (let u, find x) **K1** (substitute) **J1** (answer in x)
	TOTAL	**14 MARKS**

NO	ANSWER SCHEMES	REMARKS
5(a)	1) $f(1) = 1$ (Defined). 2) $\lim\limits_{x\to1^-}\left(x^3+2\right) = (1)^3 + 2 = 3$ $\lim\limits_{x\to1^+}\left(\dfrac{x^2+x-2}{x-1}\right) = \lim\limits_{x\to1^+}\dfrac{(x-1)(x+2)}{x-1}$ $\lim\limits_{x\to1^+}(x+2) = (1)+2 = 3$ Since $\lim\limits_{x\to1^-} f(x) = \lim\limits_{x\to1^+} f(x) = 3$, therefore $\lim\limits_{x\to1} f(x) = 3$ (Exist). 3) $f(1) \neq \lim\limits_{x\to1} f(x)$ Therefore, $f(x)$ discontinuous (not continuous) at $x = 1$.	**K1** **K1** **J1** **K1** **J1**
5(b)	$\lim\limits_{x\to1^-}\dfrac{x^2+1}{x-1} = -\infty \quad , \quad \lim\limits_{x\to1^+}\dfrac{x^2+1}{x-1} = +\infty$ The vertical asymptote is $x = 1$.	**K1 K1** **(x approaching 1 from left and right)** **J1** **K1** **(x approaching positive infinity)**

	$\lim_{x\to+\infty}\frac{x^2+1}{x-1}=\lim_{x\to+\infty}\frac{\frac{x^2}{x}+\frac{1}{x}}{\frac{x}{x}-\frac{1}{x}}=\lim_{x\to+\infty}\frac{x+\frac{1}{x}}{1-\frac{1}{x}}=+\infty$ $\lim_{x\to-\infty}\frac{x^2+1}{x-1}=\lim_{x\to-\infty}\frac{\frac{x^2}{x}+\frac{1}{x}}{\frac{x}{x}-\frac{1}{x}}=\lim_{x\to-\infty}\frac{x+\frac{1}{x}}{1-\frac{1}{x}}=-\infty$ No horizontal asymptote for $f\left(x\right)=\frac{x^2+1}{x-1}$	**K1** **(x approaching negative infinity)** **J1**
5(b)		**R1** **R1** **R1**
	TOTAL	**14 MARKS**
6 (a)	$at\,(1,1)\quad 1=\frac{a+4(1)-(1)^2}{2-a(1)-(1)^2}$ $1=\frac{a+3}{1-a}$ $a=-1$ $Thus,\, y=\frac{-1+4x-x^2}{2+x-x^2}$	**K1** **J1**
6(b)	To determine the vertical asymptote	

	$2+x-x^2=0$ $(1+x)(2-x)=0$ $x=-1, x=2$ $\lim_{x\to\infty}\dfrac{-1+4x-x^2}{2+x-x^2}$ $\lim_{x\to\infty}\dfrac{\frac{-1+4x-x^2}{x^2}}{\frac{2+x-x^2}{x^2}}$ $=1$ Horizontal asymptote y =1	**K1** **J1** **K1** **J1**
6(c)	By using long division $$\begin{array}{r} 1 \\ -x^2+x+2\overline{)-x^2+4x-1} \\ \underline{-x^2+x+2} \\ 3x-3 \end{array}$$ $\dfrac{-1+4x-x^2}{2+x-x^2}=1+\dfrac{3x-3}{2+x-x^2}$ $\dfrac{3x-3}{2+x-x^2}=\dfrac{3x-3}{(1+x)(2-x)}=\dfrac{A}{1+x}+\dfrac{B}{2-x}$ $3x-3=A(2-x)+B(1+x)$ $x=2,\ B=1$ $x=-1, A=-2$ $\dfrac{-1+4x-x^2}{2+x-x^2}=1-\dfrac{2}{1+x}+\dfrac{1}{2-x}$	**K1** **(long division)** **B1** **K1** **J1**

	$y=1-2(1+x)^{-1}+(2-x)^{-1}$ $\frac{dy}{dx}=\frac{2}{(1+x)^2}+\frac{1}{(2-x)^2}$ $When\, x=0,\ \frac{dy}{dx}=\frac{2}{(1+0)^2}+\frac{1}{(2-0)^2}=\frac{9}{4}$ $\frac{d^2y}{dx^2}=\frac{d}{dx}\left(\frac{2}{(1+x)^2}+\frac{1}{(2-x)^2}\right)$ $=\frac{-4}{(1+x)^3}+\frac{2}{(2-x)^3}$ $when\, x=0,\ \frac{d^2y}{dx^2}=\frac{-4}{(1+0)^3}+\frac{2}{(2-0)^3}$ $=-\frac{15}{4}$	**K1** **J1** **K1** **J1**
	TOTAL	**14 MARKS**
7 (a)	$f(x)=x+\frac{9}{x}=x+9x^{-1}$ $f'(x)=1-\frac{9}{x^2}$ $f'(x)=0$ $1-\frac{9}{x^2}=0$ $\frac{x^2-9}{x^2}=0$ $x^2-9=0$ $x=\pm 3$ When $x=3,\ f(3)=3+\frac{9}{3}=6,$ $x=-3,\ f(-3)=-3+\frac{9}{-3}=-6$ $\therefore$ The stationary are $(3,6)$ and $(-3,-6)$.	**K1** (find $f'(x)$ and =0) **J1** (all correct)

	$f''(x)=\frac{18}{x^3}$	**K1** (find $f''(x)$)
	When $x=3,\ f''(3)=\frac{2}{3}>0$ (minimum) $x=-3,\ f''(-3)=-\frac{2}{3}<0$ (maximum)	**K1** (apply correct theorem)
	$\therefore (3,6)$ is a relative minimum and $(-3,-6)$ is a relative maximum	**J1** (all correct)
7(b)	Given $\frac{dv}{dt}=80\pi cm^3 s^{-1}$	**B1**
	$V=4\pi h(h+4)$ $=4\pi h^2+16\pi h$ $\frac{dv}{dh}=8\pi h+16\pi$	**K1** (correct differentiate)
	$\frac{dh}{dt}=\frac{dh}{dv}\times\frac{dv}{dt}$ $=\frac{80\pi}{8\pi(h+2)}$ $=\frac{10}{h+2}$	**K1** (correct formula chain rule)
	When $h=6\ cm,\ \frac{dh}{dt}=\frac{10}{6+2}$	**K1** (substitute value)
	$=\frac{5}{4}\ cms^{-1}$	**J1**
	TOTAL	**10 MARKS**